



Predicting the mold-glass interface adhesion in thermal molding

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<https://doi.org/10.1016/j.ijmecsci.2025.110265> 

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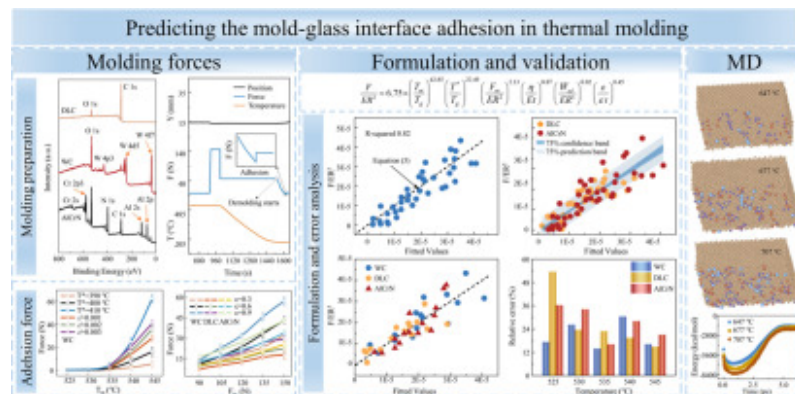
Highlights

- Developed a novel formula for rapid prediction of adhesion forces.
- Revealed how glass molding conditions and glass properties influence adhesion.
- Pinpointed molding temperature as the key factor governing adhesion.
- Discovered the atomic-level mechanisms behind adhesion behavior.

Abstract

Adhesion between a mold and a glass workpiece during demolding remains a significant challenge in precision glass molding, limiting process efficiency and product quality. Although previous studies have examined factors influencing adhesion, they are limited and predominantly qualitative, lacking a reliable framework to predict adhesion force across different mold-glass systems. This study develops a new dimensionless formulation incorporating twelve key variables to quantify the relationship between molding factors and adhesion force. The model is established through molding experiments on d-FK95 glass with three distinct mold materials, complemented by molecular dynamics simulations. The study evaluates contributions from molding parameters, glass Young's modulus, viscosity, and thermodynamic adhesion work, with atomistic interaction energy analyses at varied molding temperatures. DLC molds demonstrate superior anti-adhesion performance compared to WC and AlCrN due to their lower surface energy and amorphous structure. The model identifies molding temperature as the dominant factor, with an exponent of 42.85, followed by demolding temperature and molding force. The accuracy and applicability of the model are validated through a comparison between experimental data and fitted values for two different mold materials. The results show that the proposed model demonstrates high applicability, underscoring the critical role of precise molding parameter control in minimizing adhesion-induced defects.

Graphical abstract



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Keywords

Precision glass molding; Optical glass; Surface adhesion; Molding force; Predictive modeling

1. Introduction

Precision Glass Molding (PGM) has emerged as an important technology for the mass production of intricate glass components, offering dimensional precision and exceptional surface finish [\[1\]](#), [\[2\]](#), [\[3\]](#). Extensive research has been conducted on improving the lens profile precision and optical quality throughout a PGM process [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#). A significant challenge lies in that glass-mold adhesion during demolding occurs under the stringent processing conditions of elevated temperatures and pressures [\[8,9\]](#), because of a complex interplay of chemical and physical forces at the glass-mold interface, such as the van der Waals force, chemical bonding, and mechanical interlocking [\[10\]](#). Such interactions at the glass-mold interface could compromise the surface integrity of components, leading to detrimental effects on optical performance, increasing operational cost, and shortening the lifespan of expensive molds [\[11,12\]](#).

Addressing the glass-mold adhesion in PGM necessitates comprehensive understanding of the impacts of various factors, including the properties of the mold and glass materials [\[13\]](#), [\[14\]](#), [\[15\]](#), the process parameters [\[16\]](#), and specific demolding conditions [\[17\]](#). Specifically, mold surface roughness, chemical compatibility between mold and glass, thermal deformation properties of glass and processing temperature and pressure play vital roles [\[18,19\]](#). The preceding underscores the critical importance of an integrated approach to optimizing the PGM process.

Currently, the strategy has been on the following three main aspects: (i) controlling mold surface roughness to reduce mechanical interlocking at the contact interface, often requiring surface treatments to achieve roughness below 20 nm, meeting typical optical component produce standards [\[20\]](#); (ii) selecting coatings that can withstand the harsh conditions of the molding process while retaining their anti-adhesive properties [\[21,22\]](#), and (iii) optimizing the process parameters [\[23,24\]](#), including

temperature, pressure, and cooling rates during the PGM process to minimize adhesion on mold surfaces. For example, Li et al. [25] investigated the effect of molding parameters on surface roughness in precision glass molding, with optimized conditions achieving roughness levels suitable for high-precision optical applications. Lu et al. [26] explored the surface integrity of binderless WC processed by dry electrical discharge-assisted grinding (DEDAG), emphasizing that this method not only reduces surface roughness but also minimizes subsurface damage. Akhtar et al. [27] examined the thermal stability and anti-sticking performance of Ru–Pt multilayer coatings in high-temperature molding environments, demonstrating that the multilayer structure effectively suppresses atomic diffusion and coating degradation. Shu et al. [28] employed step-and-repeat hot imprinting to fabricate PMMA masters with continuous-profile diffractive lens arrays, achieving surface roughness suitable for optical replication. Shu et al. [29] demonstrated the use of single-point diamond turning with fast tool servo to fabricate Ni–P molds with surface roughness ($Sa < 4$ nm), ensuring minimal mechanical interlocking and high-fidelity replication of microstructures during precision glass molding. Although these strategies effectively mitigate adhesion to some extent, the influence of high temperature and pressure during molding and demolding remains unclear. Mold surface adhesion typically occurs when the interface temperature exceeds the viscosity range of $10^{8.8}$ to $10^{6.6}$ Pa.s (just above the softening point, T_s) [30,31]. This elevated temperature, usually around 80 °C beyond the glass transition temperature (T_g), well beyond T_s , inevitably leads to pronounced interface adhesion.

Despite some advancements, challenges persist in the practical implementation of anti-sticking strategies. Achieving ultra-smooth surfaces on complex molds remains a significant hurdle. While contemporary polishing techniques can easily achieve nanometer-scale roughness on flat molds, intricate mold forms require more complicated and costly machining processes like single-point diamond turning [32,33]. As a result, the most common approach is to apply anti-sticking coatings. Three main options are available: ceramic coatings, precious metal coatings, and carbon films [[34], [35], [36]]. While effective in reducing adhesive challenges, ceramic coatings (e.g., TaN, TiAlN, CrWN) [37,38] show poor thermal stability after multiple molding cycles. At temperatures exceeding 500 °C [39], metallic elements are prone to oxidation, leading to glass adhesion and coating deterioration. Precious metal coatings (e.g., Pt–Ir, Ir–Re and Ru–Pt) offer superior performance but are prohibitively expensive, although the Pt–Ir coating system is still the most widely used due to its high oxidation and corrosion resistance. Carbon films (DLC) [40,41], though providing good surface cleanliness, have limited applications due to their vulnerability to oxidation at high temperatures. Zhang et al. [42] demonstrated a nano-graphitic carbon coating via APCVD on Si molds, offering hydrophobicity and thermal stability, addressing carbon film oxidation vulnerabilities. Yu et al. [43] optimized heat-treated Ni–P coatings to enhance hardness and thermal stability, aligning with ceramic coatings' mechanical reinforcement but highlighting their

oxidation-prone nature at high temperatures. Xie et al. [44] applied Re–Ir alloy coatings to Ni–P molds to suppress phosphorus diffusion and friction, validating precious metal coatings' efficacy despite cost barriers. Huang et al. [45] characterized CrWN coatings for oxidation resistance and wear performance, yet noted thermal stability limitations akin to other ceramic systems. Kim et al. [46] deposited SiC on graphite via low-temperature chemical vapor deposition, improving anti-adhesion through carbon-rich layers while leveraging graphite's lubrication. Additionally, the efficacy of the coatings declines over time due to probable deterioration and the danger of glass surface contamination [47,48]. Therefore, understanding the adhesion behavior between glass and mold surfaces becomes essential for evaluating the long-term performance of these coatings.

To evaluate glass adhesion on mold surfaces, experiments have conceptualized interface adhesion strength as the peak reverse tensile force at the moment of glass-mold interface separation [49,50]. Studies have shown that under constant molding temperature, factors such as molding force and dwell time significantly influence adhesion force. Zhou et al. [51] investigated temperature and rate-dependent debonding behaviors of glass-mold interfaces using a probe tack test, revealing three debonding types and quantifying adhesion forces. Zhu et al. [52] analyzed the role of normal stress and pressing force in ChG glass–mold interfacial adhesion through cylindrical compression tests and finite element simulations, revealing that stress concentration at glass edges enhances physical adhesion, while optimized spherical preforms reduce normal stress and suppress adhesion forces. These studies collectively highlight the methods used to assess adhesion strength, revealing how molding parameters affect adhesion. However, adhesion behavior is not solely determined by process parameters; factors such as mold material properties and glass thermal deformation also play a significant role. The combined effects of these factors, accumulated throughout the heating and cooling stages, add significant complexity to the demolding process, making it challenging to establish direct cause-and-effect relationships. Despite the important data provided by these experimental studies, research in this area remains limited in both number and scope.

While these experiments provide valuable insights into macroscopic adhesion behavior, they fail to reveal the underlying mechanisms that govern adhesion. To address this issue, recent efforts have introduced numerical simulations, which allow for analysis of adhesion at the atomistic level [[53], [54], [55]]. Molecular dynamics simulation (MD) method has shown to be effective in other research areas for understanding the interface interaction mechanisms, especially when chemical reactions at the interface are not involved [[56], [57], [58]], as is the case with the glass-mold interaction in PGM. Srivastava et al. [59] demonstrated utility of MD method in nanoscale tribology by analyzing lubricant molecular arrangements and their impact on friction at metal surfaces. Yang et al. [60] applied MD to microinjection molding interfaces, revealing strong van der Waals interactions at COC—Ni interfaces and proposing fluorine doping to reduce adhesion. Liu et al. [61] compared polymer-metal

interfaces (PPS/PA6/iPP with Al/Cu) in nano-molding, highlighting polymer polarity and substrate material as critical factors for interfacial anchoring. Collectively, these studies underscore role of MD simulations in understanding interfacial phenomena [62,63]. In glass molding, the primarily focus of MD simulations has been on understanding of properties of glass surfaces [64,65] and developing potential models for moldable glass [[66], [67], [68]], while few have addressed the issue of adhesion. To our knowledge, no investigation has been on the glass-mold adhesion aiming to minimize mold surface defects and enhance the optical quality of glass.

As a continuation of previous research of glass molding, this study aims to systematically investigate the adhesion mechanism in PGM through both experimental and computational approaches. To achieve this, the arrangement of this paper is as follows: Section 2 introduces the experimental and MD simulation methodologies, detailing the molding tests and parameter settings, as well as the computational framework used to model interfacial adhesion at the atomic scale. Section 3 presents the experimental and simulation results, analyzing the influence of molding and demolding parameters on adhesion forces for different mold surfaces. A dimensionless formulation is developed to quantitatively describe the relationship between key influencing factors and adhesion force. Additionally, the simulated adhesion process and interaction energy are examined to provide deeper insights into adhesion mechanisms at the interface. Finally, the conclusions and implications for refining PGM processes are summarized in Section 4.

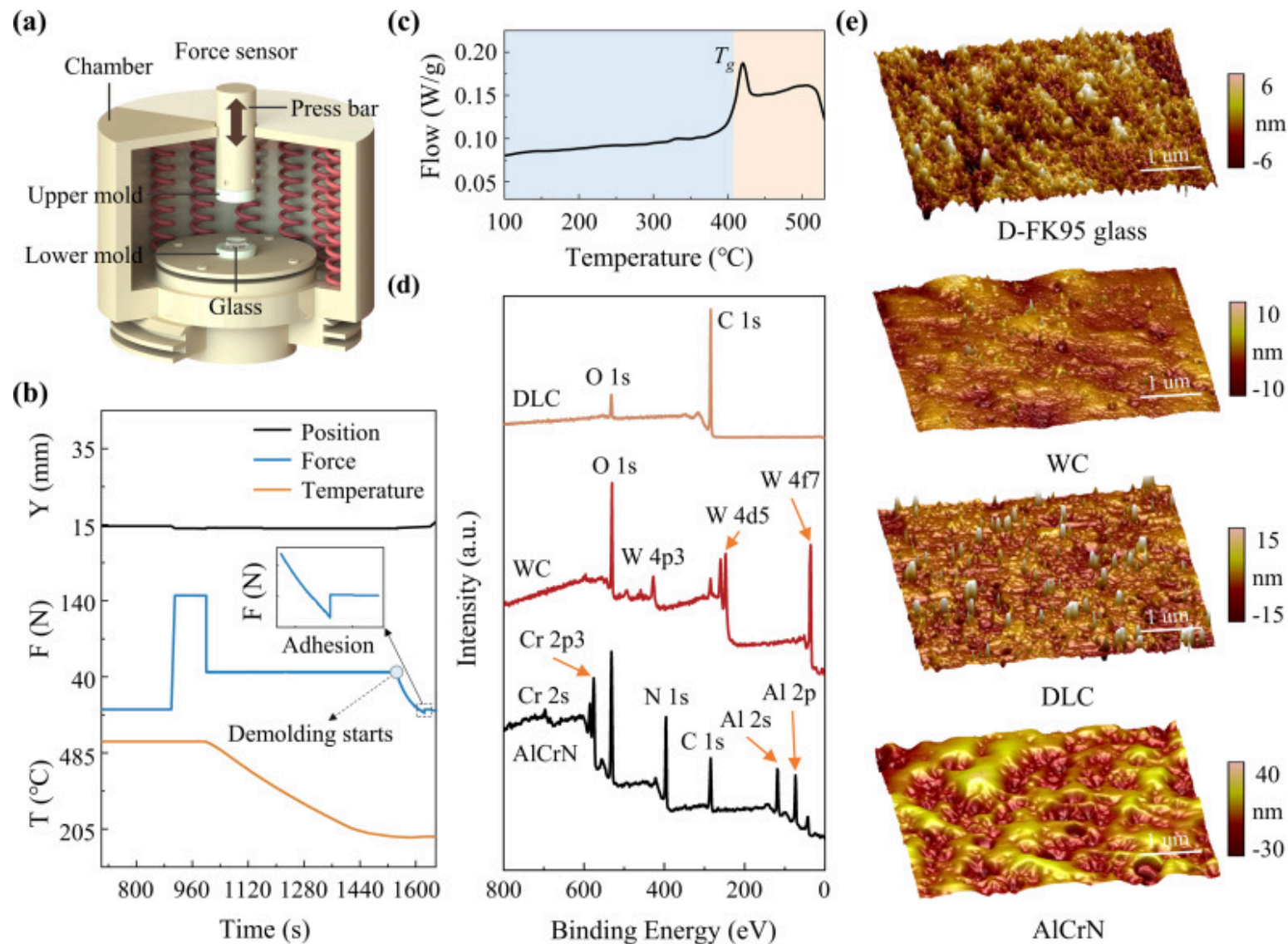
2. Methodology

This section describes the experimental and computational methodologies adopted in this study. First, the experimental setup and procedures for measuring demolding adhesion are introduced, detailing the key parameters and conditions that influence adhesion at the glass-mold interface. Then, the MD simulation framework is presented, including the model construction, potential functions, and boundary conditions. The combined approach enables a comprehensive understanding of adhesion behavior by integrating empirical observations with atomistic insights.

2.1. Experiment

To assess demolding adhesion, the adhesion force during the separation at the glass-mold interface was first measured experimentally. The demolding process involved pressing two flat-ended molds against a d-FK95 cylindrical glass sample (provided by NHG Glass Co. Ltd.), maintaining the pressure for a set duration, and then retracting the mold at a reducing

temperature. This operation was conducted within a standard high-temperature chamber (Rtec Instrument CA95131) on the MFT-5000 apparatus, as depicted in [Fig. 1\(a\)](#). The sample setup adhered to the specifications outlined in a previous study [[10](#)]. The precision of the entire apparatus was paramount, achieving a displacement resolution of $0.25\text{ }\mu\text{m}$ along the Z-axis, a temperature resolution of $0.1\text{ }^{\circ}\text{C}$, and a force transducer capable of measuring up to 200 N. The force sensor was attached to the upper mold through the press bar with the resolution of 6 mN, thereby ensuring the reliability of the experiment. The temperature, load, and mold position were meticulously monitored and regulated at 10 Hz throughout the compression tests.



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Fig. 1. Experimental preparation. (a) Overview of the experimental setup. (b) Representative molding process at 525 °C. The key properties of the glass and molds are shown in (c–e). (c) DSC thermal curve indicating the glass transition temperature (T_g). (d) XPS analysis of three mold materials. (e) Initial surface roughness.

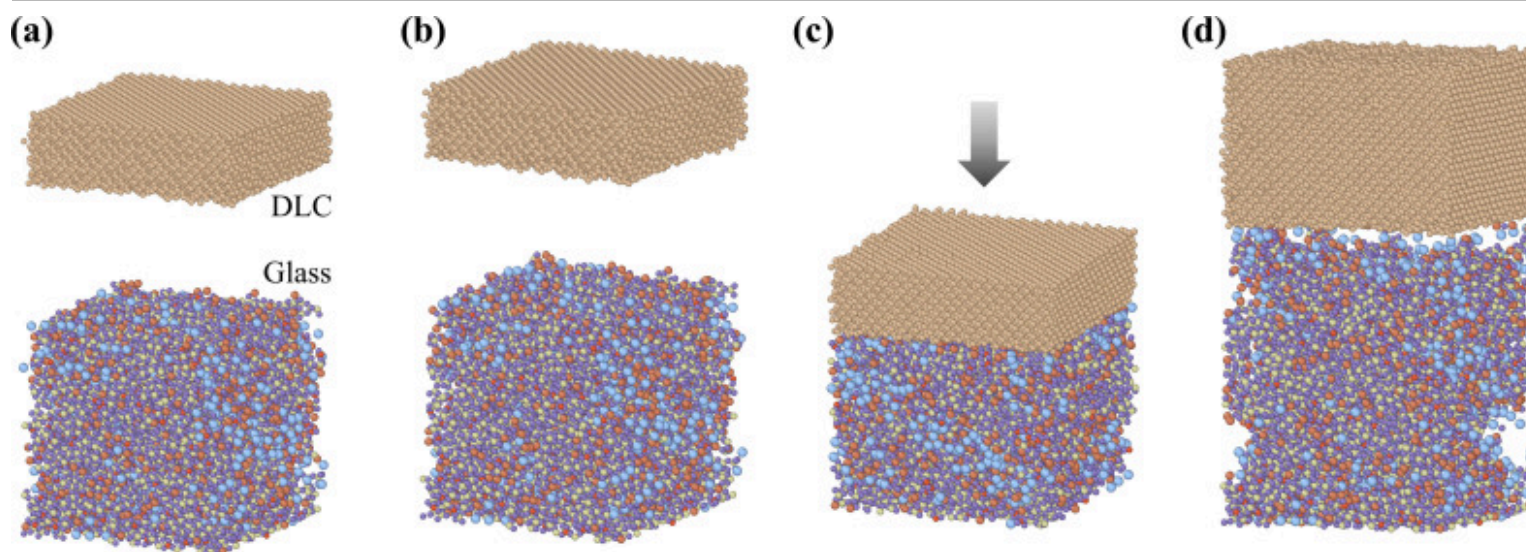
The representative experimental process parameters at a molding temperature of 525 °C are detailed in Fig. 1(b). After soaking at 525 °C for 900 s, the upper mold was lowered down to apply a pressure of 150 N for 120 s. The system was then cooled to the demolding temperature near T_g with a holding force of 60 N and a cooling rate of 13 °C/min. The demolding process began with the upper mold ascending at 10 μm/s until reached the predetermined separation distance of 2 mm. When initiating the demolding process, the force detected by the sensor transitions from positive to negative values. The recorded negative force corresponds to the adhesive forces during demolding, as shown in the inset of Fig. 1(b). The adhesion force is defined as the absolute value of the measured negative peak, which occurs just before the interface separates and the force returns to 0 N. The molding temperatures (T_m) ranged from 525 to 545 °C, demolding temperatures (T^*) from 390 °C to 430 °C, and molding forces (F_m) from 90 N to 150 N. The strain (ε) varied at 0.3, 0.6, 0.9 and strain rate ($\dot{\varepsilon}$) at 0.001, 0.002 and 0.003. For each test group, the studied variables were adjusted while all other parameters were held constant. Specifically, for tests varying T_m or T^* , F_m was kept constant throughout the pressing duration for each coating. In tests with varying F_m , T_m and T^* were fixed until the same strain was reached. For tests varying the pressing strain rate ($\dot{\varepsilon}$), T_m and T^* were held constant until reached the same pressing force and maintained until the pressing time elapsed.

The diameter and thickness of the cylindrical optical glass d-FK95 were 8.0 mm and 3.0 mm, respectively. The molding temperatures were set to align with the standard molding technology, and confirmed by differential scanning calorimetry tests (Fig. 1(c)). The glass transition temperature was measured to be 410 °C. Fig. 1(d) presents the XPS analysis of three mold coatings, revealing their elemental composition and chemical states. The DLC spectrum exhibits a dominant C 1 s peak at 284.8 eV, attributed to C–C bonds, indicating the presence of sp^2 and sp^3 hybridized carbon structures. Additionally, a minor O 1 s peak at 532.8 eV suggests the presence of surface-adsorbed oxygen or possible C–O bonding, likely due to slight oxidation or contamination from ambient exposure. For the WC coating, multiple tungsten-related peaks are observed. The W 4f7/2 peak at 31.9 eV corresponds to W–C bonding, confirming the formation of tungsten carbide as the primary phase. A peak at 35.8 eV, which can be separated from the W 4f5/2 peak near 34.62 eV, suggests the presence of a minor amount of WO_3 , likely from limited surface oxidation during sample exposure to air. This indicates that the WC coating is in good initial condition, with only superficial oxidation that is common for such materials. For the AlCrN coating, the elemental composition is estimated to be $Al_{42}Cr_{16}N_{42}$, based on high-resolution XPS spectra. A strong Cr 2p3/2 peak appears at 574.3 eV, characteristic of metallic Cr bonding, while the Al 2p peak at approximately 72.9 eV indicates the presence of Al–N bonding in the nitride phase. The N 1 s peak at 396.53 eV corresponds to metal nitrides (Al–N and Cr–N), confirming the formation of a stable nitride structure. The surface roughness of both the glass and molds was examined by atomic force microscopy (AFM) in Fig. 1(e). The roughness (R_a)

of the glass sample was 1.26 nm, while that of the mold surfaces was 0.97, 3.17, and 8.73 nm for WC, DLC, and AlCrN, respectively, which are below 10 nm and are acceptable due to limited roughness not to cause an evident impact on adhesion force.

2.2. MD analysis

An instant observation of atomistic deformation in experiments is challenging. MD simulations were therefore carried out for understanding the underlying adhesion mechanisms. To be aligned with the experimental analysis, glass N-BK7 was again used in the MD system with LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator). To replicate the glass structure accurately, the partial structure factors via the Fourier transformation of pair distribution functions (PDFs) were calculated to ensure alignment with the empirical neutron scattering data. Subsequently, nanoindentation simulations were conducted to examine the deformation characteristics of glass, with a 5 Å fixed boundary layer and an adjacent 5 Å thermostatic layer. The DLC model for molding simulation was subsequently constructed. Interface adhesion was analyzed throughout PGM process consisting of heating, soaking, pressing, holding and interface separation, as shown in Fig. 2 at the molding temperature of 647 °C. The sequential process is as follows: initial relaxed model (Fig. 2(a)), structure after soaking (Fig. 2(b)), structure during pressing (Fig. 2(c)), and interface during separation (Fig. 2(d)) where glass adheres to the DLC surface.



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Fig. 2. Simulated DLC-glass interface model throughout the molding process. (a) Initial contact interface. (b) Relaxed model after soaking. (c) Interface during the pressing stage. (d) Model during the demolding phase.

The details of the simulation are presented in [Table 1](#). For the glass structure, interactions were described by the modified two-body Buckingham potential [69]. A heating-quenching process was employed to prepare the bulk glass sample, involving cooling from 800 to 25 °C. During quenching, the stress tensor was maintained at zero using the Parrinello-Rahman barostat, and the temperature was regulated by the Nosé-Hoover thermostat. After preparing the glass, the initial contact interface was formed by placing the upper DLC model that was initially 40 Å above the surface of the glass sample. The potential for DLC is described by the Tersoff potential with a cutoff of 11 Å [70], while the DLC-glass atomic interactions are modeled using a combination of Lennard-Jones and Coulombic pairwise interactions.

Table 1. Detailed parameters for MD simulation of interface adhesion.

System	DLC-(N-BK7)
Timestep (fs)	0.1
Dimensions (Å)	62.8 × 62.8 × 192.8
Number of atoms	23,440
Boundary conditions	Periodic boundary condition applied to all surfaces except the top surface of the mold, the bottom surface of glass and the glass-mold interface
Molding temperature (°C)	647/677/707
Total glass strain (%)	40
Pressing and separation speed (Å/ps)	5
Heating rate (°C/ps)	5
Cooling rate (°C/ps)	5

System	DLC-(N-BK7)
Holding time (ps)	10

The interfacial interaction force was characterized by van der Waals and electrostatic interaction energy, i.e.,

$$E_{\text{interaction}} = E_{\text{vdw}} + E_{\text{ele}} = 4\epsilon \left[(\sigma/r)^{12} - (\sigma/r)^6 \right] + (q_i q_j) / r \quad (r < r_c) \quad (1)$$

where ϵ is the interaction constant, σ is the equilibrium distance between two atoms, r is the instant distance between two atoms, r_c is the cutoff distance, and q_i and q_j represent the atom charges. The cutoff distance for both Lennard-Jones potential and Coulombic pairwise interaction was set to be 1.1 nm.

In the MD modelling, the bottom layer of 5 Å was fixed in the space as the boundary atoms to eliminate its rigid body motion. The region in the glass adjacent to this fixed layer, also with a thickness of 5 Å, was treated as a thermostatic layer. The energy of the Newtonian atoms was controlled by the NVT ensemble. The mold atoms were pushed down towards the glass surface by applying a uniform displacement on the layer top atoms.

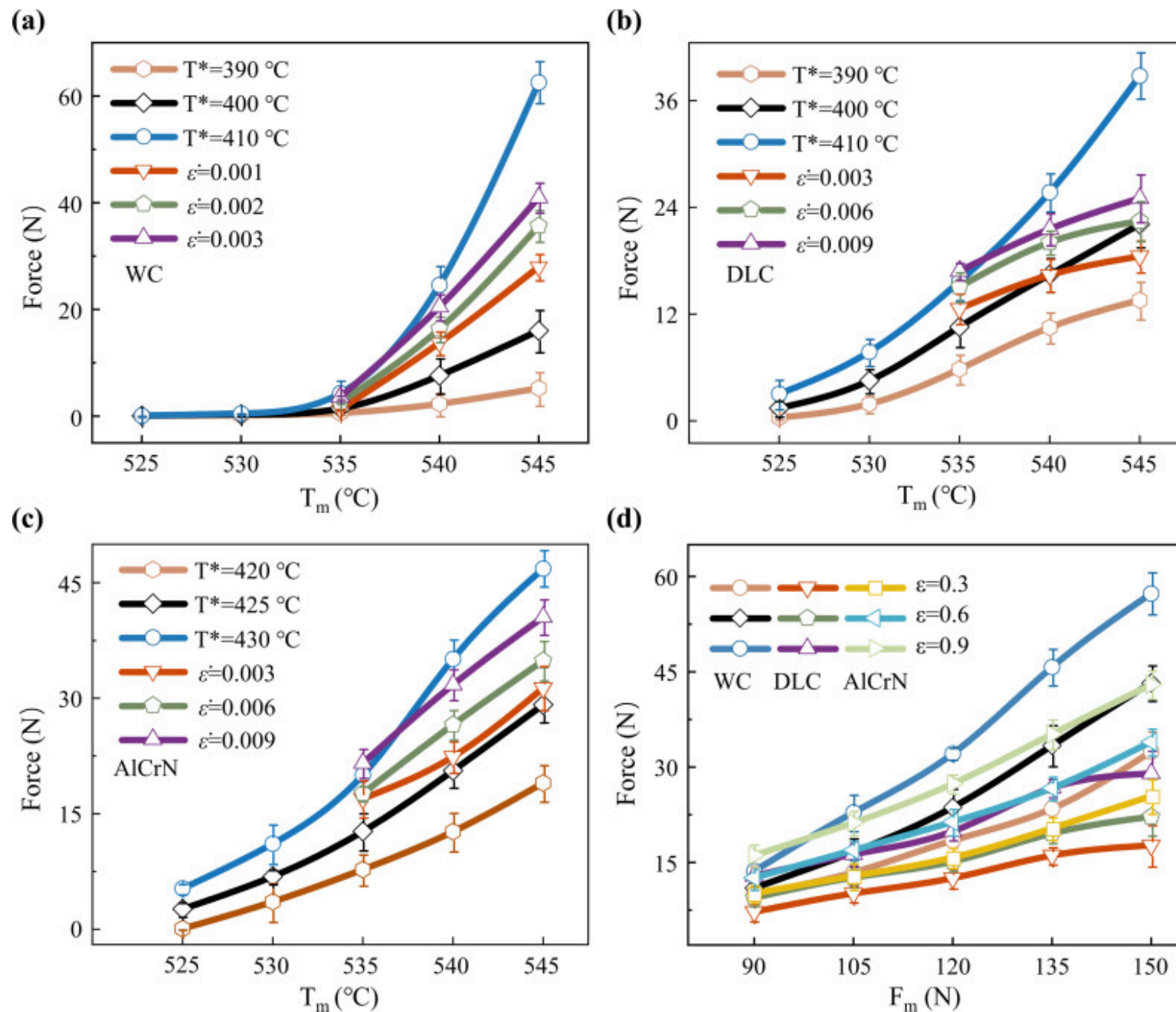
3. Results and discussion

In this section, a quantitative adhesion force model was developed and its validity was verified. Furthermore, the atomic adhesion behavior on DLC molds at different molding temperatures was analyzed. The adhesion force mechanism was explored through key parameters and interface interaction energies.

3.1. Experimental results

Adhesion forces variation with the three mold materials under various molding conditions are shown in Fig. 3. Each test was repeated for three times, and the average data were presented with error bars. Fig. 3(a) to (c) show that greater molding temperature (T_m) and demolding temperature (T^*) contribute to higher adhesion forces, causing severer surface adhesion. At the molding temperatures of 525 °C and 530 °C, the adhesion force for the WC mold is almost negligible. This is attributed to the high viscosity and restricted fluidity of the glass, which maintains the glass-mold interface contact largely unaltered, resulting in minimal adhesion forces. In contrast, for the AlCrN mold, the adhesion force increased from 5.3 N to 11.1 N, possibly due to the

PVD deposition process used for the coating, as shown in Fig. 3(c). This method typically induces micro-defects such as pinholes and particles of varying sizes with uneven distribution [71]. These defects create additional contact areas, increasing the likelihood of glass adhesion to the surface. For the DLC mold surface, its amorphous characteristics result in lower adhesion forces compared to the AlCrN mold, as shown in Fig. 3(b). Fig. 3(d) shows the adhesion forces of the three mold materials under four different molding forces and three strain levels. The results indicate that lower molding forces (F_m) and smaller strains (ϵ) lead to reduced adhesion forces, suggesting that molding force is another potentially significant factor. However, the effect of strain rate ($\dot{\epsilon}$) appears comparatively less important. These findings demonstrate that adhesion force is a multifaceted outcome influenced by various molding parameters.

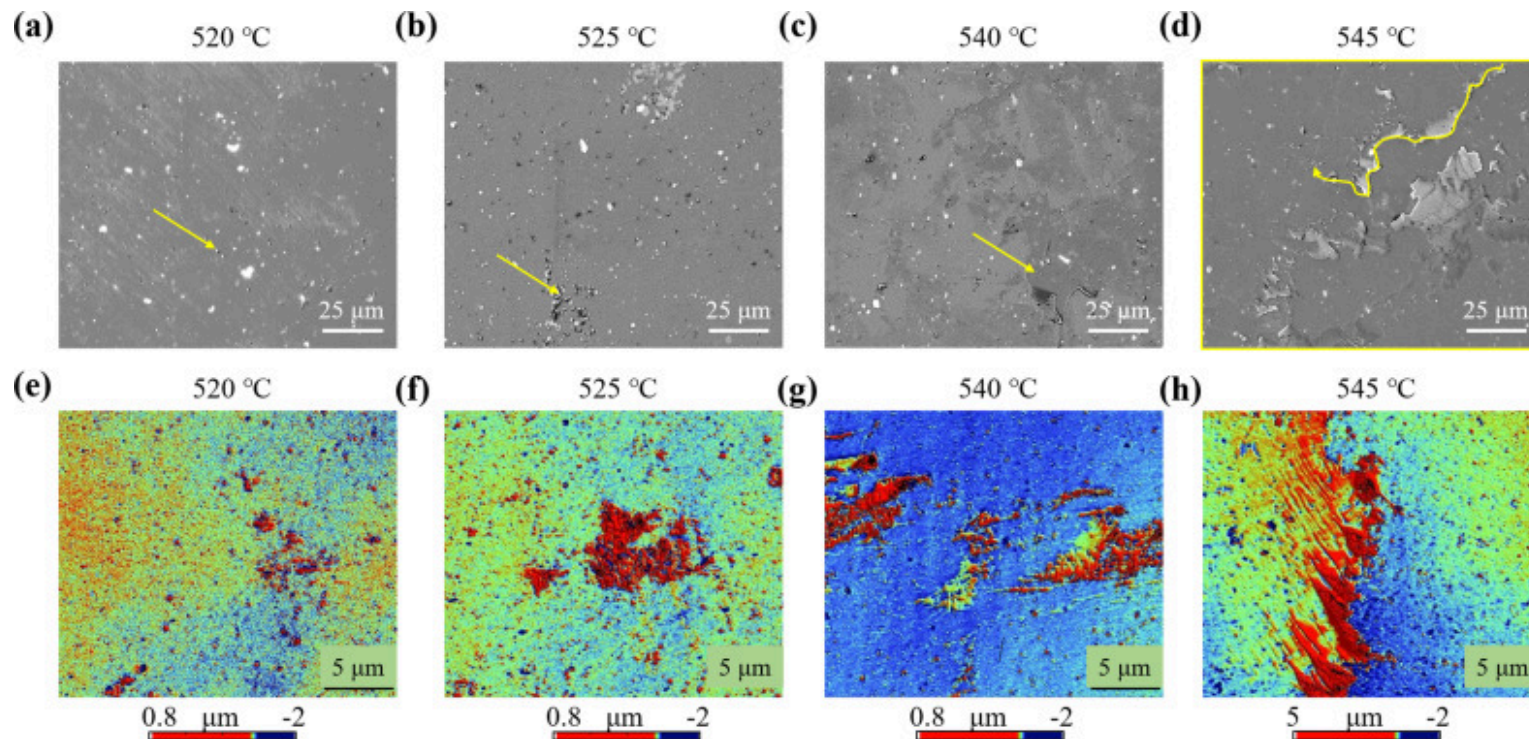


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Fig. 3. Adhesion forces under different molding conditions for WC, DLC and AlCrN molds. (a-c) Effects of molding temperature (T_m) and demolding temperature (T^*). (d) Influence of molding force (F_m) and strain (ε).

An investigation into the mold surface morphology after adhesion provided insights into the distribution characteristics of glass adherence. Since the adhesive morphology of the three molds exhibits similar characteristics, we use the DLC coating as a representative example to illustrate the morphological features and their evolution process, as shown in Fig. 4. Fig. 4(a–d) reveal a transition in surface morphology, progressing from isolated dot-like glass deposits to extensive linear accumulations, culminating in planar residues spanning tens of micrometers with a height of $<1\ \mu\text{m}$ (as indicated by yellow arrows and dotted lines). The corresponding adhesion forces increased from 0.67 N to 4.10 N, 21.94 N, and finally 38.26 N as the molding temperature gradually increased, corresponding to the adhesion morphologies shown in Fig. 4(a–d). The surface roughness (R_a) of the molds after a single molding cycle was found to be 46.47, 25.62, 19.28, and 12.74 nm, for 545 °C, 540 °C, 525 °C, and 520 °C, respectively, as shown in Fig. 4(e–h). Such a substantial increase indicates significant mold surface contamination by adhered glass, likely impairing the effectiveness of the molds in subsequent molding cycles. Specifically, the surface roughness in the range from 540 to 545 °C increased more than tenfold from the original DLC mold surfaces. This effect is primarily attributed to the expanded interface contact area and the low competence of DLC molds at high temperatures, thereby intensifying the surface adhesion. The above variation in adhesion force and morphology characteristics highlights the significant impact of molding parameters on demolding adhesion. However, demolding adhesion, as a phenomenon of contact mechanics, is influenced by a broader range of factors. Therefore, a comprehensive study of the adhesion mechanism is essential to fully understand the issue.



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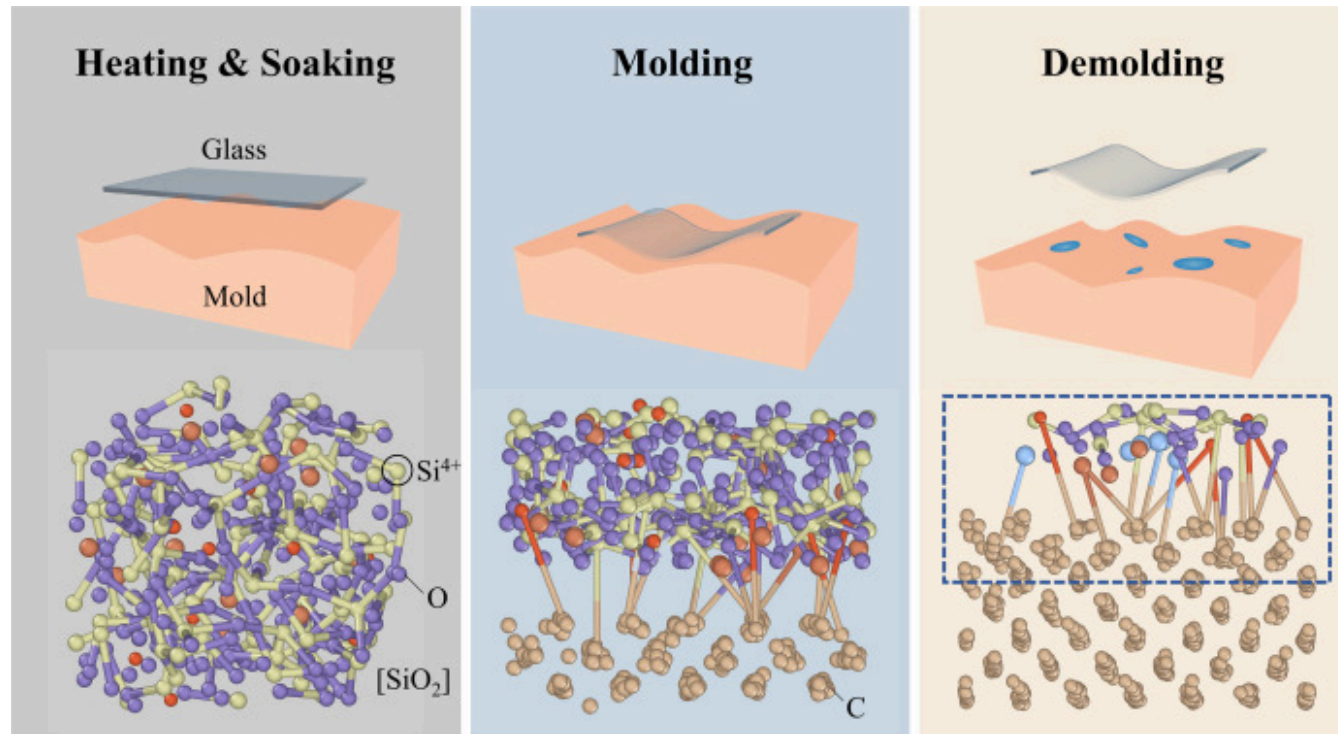
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Fig. 4. Comparative analysis of surface adhesion of DLC molds at molding temperatures ranging from 520 to 545 °C. (a-d) SEM images showing surface morphologies. (e-h) 3D profiles of adhered characterizations.

3.2. Critical factors

Understanding the adhesion mechanism at the glass–mold interface is inherently complex due to the interplay of multiple factors at different stages of the process [[72], [73], [74]]. As illustrated in Fig. 5, during the heating and soaking phase, the glass primarily exhibits a SiO₂ network structure. Upon molding, physical bonding forms between the glass and the DLC mold surface [10]. In addition, non-bonding interactions arise due to the affinity between the mold surface atoms and the glass atoms, further contributing to interfacial adhesion and ultimately leading to adherence during demolding. The dominant governing factors vary across the different stages: in the heating and soaking phase, the thermodynamic work of adhesion [75] and the geometric

characteristics of the glass play a critical role, influencing material selection and interface properties. During molding, adhesion is strongly affected by processing conditions and the mechanical response of the glass, which depend on its viscosity and Young's modulus at elevated temperatures [76,77]. Finally, during demolding, temperature critically dictates the interfacial separation behavior.



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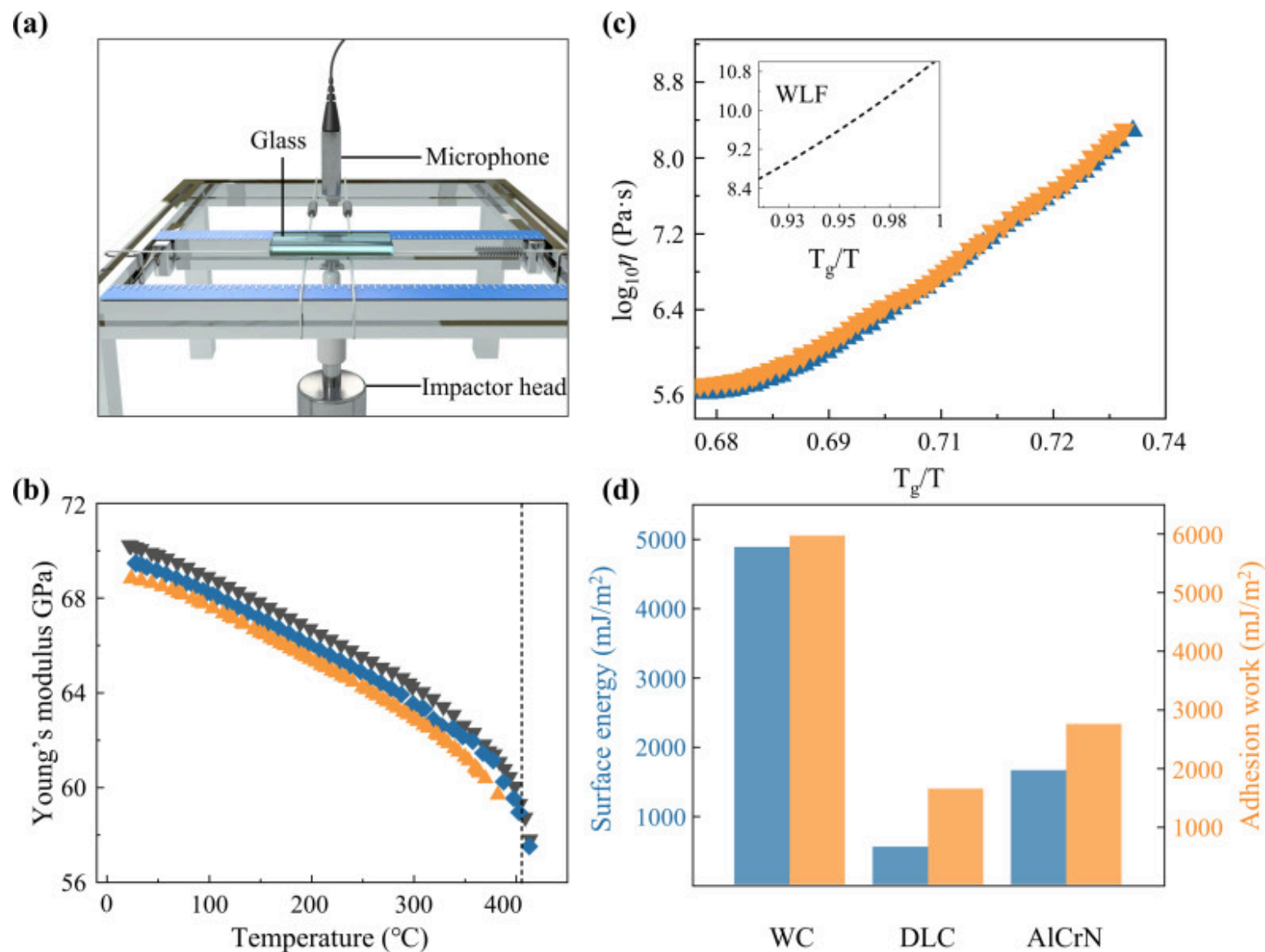
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Fig. 5. Adhesion analysis at the DLC mold–N-BK7 glass interface using dimensionless analysis and molecular dynamics simulations.

To provide a comprehensive understanding of these interdependent factors, a dimensionless analysis framework integrates multiple variables, while MD simulations capture interface interactions during separation. In the following, we examine the key controlling parameters—Young's modulus, glass viscosity, and thermodynamic work of adhesion—alongside experimental validation.

3.2.1. Young's modulus

The Young's modulus of the glass determines its capacity to withstand and recover from stresses during the thermal deformation in molding [[78], [79], [80]]. Higher modulus indicates increased stiffness, which reduces the deformability of the glass around the glass-mold interface, thereby decreasing adhesion. The Young's modulus of the d-FK95 glass sample was assessed with an IMCE-RFDA device, which employs automated impulse excitation for measurements inside a high temperature furnace and the Fourier analysis of the sound wave, the frequency, as well as the dimensions and mass of the specimen, is used to calculate the Young's modulus according to the ASTM E1876–22 standard, as shown in Fig. 6(a). The $3 \times 20 \times 60$ mm glass plates were grounded to ensure surface flatness and the testing heating rate of 5 °C/min with protection of argon inside chamber.



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Fig. 6. Critical factors influencing mold surface adhesion. (a) Measurement unit of glass modulus. (b) Temperature-dependent curve of Young's modulus. (c) Temperature-dependent glass viscosity. (d) Surface energy of mold materials and interface

adhesion work when paired with d-FK95 glass.

Data spanning a temperature range of 25–410 °C, presented in Fig. 6(b), demonstrate a noticeable decrease in Young's modulus from 69 GPa to 57 GPa as approaching the glass transition temperature (T_g), highlighting the gradual softening of glass with the temperature rise. Separated by T_g temperature, at stage I, the Young's modulus decreases linearly with temperature, similar to observations in crystalline material. It shows that glass atoms remain inherent atomic structure, and such a trend comply with the Debye-Gsrüneisen (DG) model. Reaching to T_g temperature, the glass specimen starts to soften due to ability to access large amount of free volume and structural relaxation, which explains the rapid decline in modulus. As the temperature keeps rising, the glass tends to be easier to viscously flow, and the modulus changes faster until the glass can't hold its original shape.

3.2.2. Viscosity

In addition to Young's modulus, glass viscosity governs the flow characteristics of the material [81] and its resistance to deformation over time, particularly at elevated temperatures in PGM. To elucidate the temperature-dependent behavior of glass viscosity, measurements were conducted using the parallel plate viscometer (PPV) technique [82,83], integrated with the Williams-Landel-Ferry (WLF) two-parameter model [84]. The compression test involved a cylinder-shaped glass specimen, subjected to uniaxial compression between two parallel plates of high stiffness. The specimen height changed in response to the variations in temperature and time. Thus, the viscosity could be expressed as [85]:

$$\eta = \frac{2\pi F h^5(t)}{3V(2\pi h^3(t)+V)} \left(\frac{dh(t)}{dt} \right)^{-1}, \quad (2)$$

where F is the loading force, V denotes the sample volume, and $h(t)$ is the instantaneous length of the sample.

The experiments were conducted within a temperature range of 540 °C to 610 °C based on previous molding settings. Under a constant compression force of 1 N, the temperature of the glass sample was increased at the rate of 10 °C/min. The results were shown in Fig. 6(c) in Angell's plot. The viscosity at 560 °C was approximately $10^{8.2}$ Pa·s, and decreased as the microstructure continued to relax, reaching a minimum viscosity of $10^{5.7}$ Pa·s around 602 °C, after which it remained at a minimum value. The calculated viscosity was further used to estimate the glass viscosity near the demolding temperature using the WLF model. The WLF model provides an empirical yet theoretically substantiated description of viscosity (η) as a function of temperature (T), especially near the glass transition temperature (T_g). The derived viscosity near the demolding temperature ranged from $10^{8.4}$ to

$10^{10.8}$ Pa·s, as presented in the inset of Fig. 6(c). This provides a detailed understanding of how viscosity changes across the temperature range of 404 °C to 450 °C, enabling a direct comparison of empirical data and theoretical insights.

3.2.3. Thermodynamic adhesion work

Apart from the mechanical properties of the glass, the thermodynamic work of adhesion also significantly influences the strength of demolding adhesion [86]. By employing Dupre's formula [87], the adhesion work between glass and mold materials was quantitatively expressed in terms of the surface energy of the mold surface and optical glass, as well as the interfacial interaction energy, which was evaluated using the macroscopic atom model [88]. The calculated results in Fig. 6(d) show that WC has the highest surface energy of 4887.14 mJ/m², while DLC has the lowest surface energy of 547.05 mJ/m². The adhesion work with the glass follows a similar trend, at the values of 1640.06, 2745.22, and 5970.21 mJ/m² for DLC, AlCrN and WC, respectively. These findings align with the experimental adhesion forces trend in Fig. 3(a), thus confirming the superior anti-adhesion properties of DLC molds.

3.3. Dimensionless formulation

The above examination on the critical factors offered a comprehensive understanding of adhesion in glass molding. However, it remains qualitative and does not provide their direct quantitative relationships with adhesion force. To address this problem, a dimensionless formulation was carried out below [[89], [90], [91], [92], [93]] to enable a more effective comparison across different influencing factors. The twelve variables involved, along with their respective dimensions, are presented in Table 2. Following the Buckingham PI theorem, the corresponding dimensionless formulation can be derived as

$$f\left(\frac{F}{ER^2}, \frac{T_m}{T_g}, \frac{T^*}{T_g}, \frac{F_m}{ER^2}, \frac{\eta}{Et}, \frac{W_{ad}}{ER^3}, \frac{\varepsilon}{\varepsilon \cdot t}\right) = 0 \quad (3)$$

Table 2. Physical quantities for dimensional analysis: symbols, meanings and dimensions.

Symbol	Physical meaning	Dimensions
F	mold-glass adhesion force	MLT ⁻²
T _m	the molding temperature	Θ

Symbol	Physical meaning	Dimensions
T^*	demolding temperature	Θ
T_g	glass transition temperature	Θ
F_m	the molding force	MLT^{-2}
η	glass viscosity at molding temperatures	$ML^{-1}T^{-1}$
W_{ad}	interface adhesion work	ML^2T^{-2}
$\dot{\epsilon}$	the strain rate of glass height during molding	T^{-1}
ϵ	the total strain of glass height	1
E	glass Young's modulus at the demolding temperature	$ML^{-1}T^{-2}$
R	initial glass cylinder radius	L
t	duration at the molding temperature	T

The dimensionless groups in Eq. (3) can be rearranged as

$\Pi_1 = \frac{F}{ER^2}$ represents the dimensionless adhesion force during the demolding;

$\Pi_2 = \frac{T_m}{T_g}$ is the ratio of the molding temperature to the demolding temperature;

$\Pi_3 = \frac{T^*}{T_g}$ indicates the ratios of molding temperature to the demolding temperature;

$\Pi_4 = \frac{F_m}{ER^2}$ is the effect of molding force on the adhesion;

$\Pi_5 = \frac{\eta}{Et}$ indicates the effect of glass viscosity at the demolding temperature;

$\Pi_6 = \frac{W_{ad}}{ER^3}$ is the effect of thermodynamics adhesion work;

$\Pi_7 = \frac{\varepsilon}{\varepsilon \cdot t}$ represents the effect of strain and strain rate.

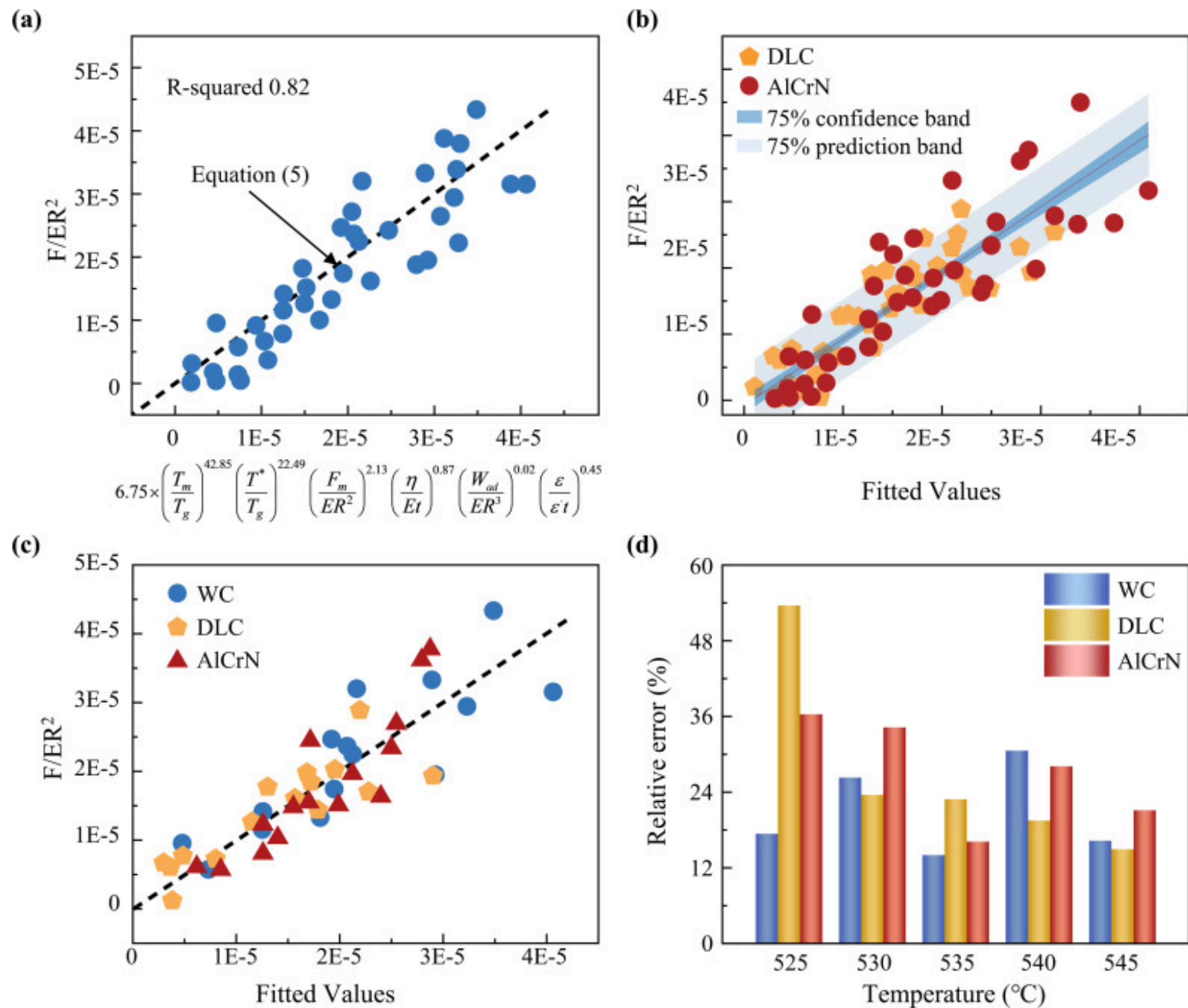
By applying the power law, Eq. (3) can be described as

$$\frac{F}{ER^2} = \kappa \times \left(\frac{T_m}{T_g}\right)^{k_1} \left(\frac{T^*}{T_g}\right)^{k_2} \left(\frac{F_m}{ER^2}\right)^{k_3} \left(\frac{\eta}{Et}\right)^{k_4} \left(\frac{W_{ad}}{ER^3}\right)^{k_5} \left(\frac{\varepsilon}{\varepsilon \cdot t}\right)^{k_6}, \quad (4)$$

where coefficients κ and k_i ($i = 1, \dots, 6$) are constants that can be determined through multiple linear regression of the experimental data for WC mold shown in Fig. 3. At a confidence level of 81.56 %, Eq. (4) became:

$$\frac{F}{ER^2} = 6.75 \times \left(\frac{T_m}{T_g}\right)^{42.85} \left(\frac{T^*}{T_g}\right)^{22.49} \left(\frac{F_m}{ER^2}\right)^{2.13} \left(\frac{\eta}{Et}\right)^{0.87} \left(\frac{W_{ad}}{ER^3}\right)^{0.02} \left(\frac{\varepsilon}{\varepsilon \cdot t}\right)^{0.45}, \quad (5)$$

where the unit of the parameters are in SI. Fig. 7 compares the predictions from Eq. (5) with experimental measurements for the WC mold and extends the analysis to DLC and AlCrN molds for validation. In Fig. 7(a), the model demonstrates a satisfactory prediction of the experimental measurements over a wide range of dimensionless adhesion force (F/ER^2) values for WC molds, with a fitting R-squared of 0.82. The sensitivity analysis indicates that adhesion force (F) is positively correlated with the molding temperature (T_m), demolding temperature (T^*), molding force (F_m), and glass viscosity (η), meaning that reducing these parameters can effectively lower adhesion force. The ranking of their influence from highest to lowest is $T_m > T^* > F_m > \eta > \varepsilon > \varepsilon \cdot t$, with exponents of 42.85, 22.49, 2.13, 0.87, 0.45, and 0.02, respectively, where molding temperature (T_m) plays the most significant role, consistent with previous literature [94]. The model was then applied to DLC and AlCrN molds, and the fitting reliability was assessed with a 75 % confidence band and prediction band in Fig. 7(b), yielding an overall R^2 of 0.74, with DLC exhibiting better fitting accuracy.



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Fig. 7. Adhesion force model development and validation. (a) Model construction using WC mold data, compared with experimental measurements (circular symbols). (b) Model fitting results for DLC and AlCrN molds, including confidence and prediction bands. (c) Original adhesion force data for all three molds as a function of molding temperature. (d) Average relative error of adhesion forces for the three molds at different molding temperatures, derived from the data in (c).

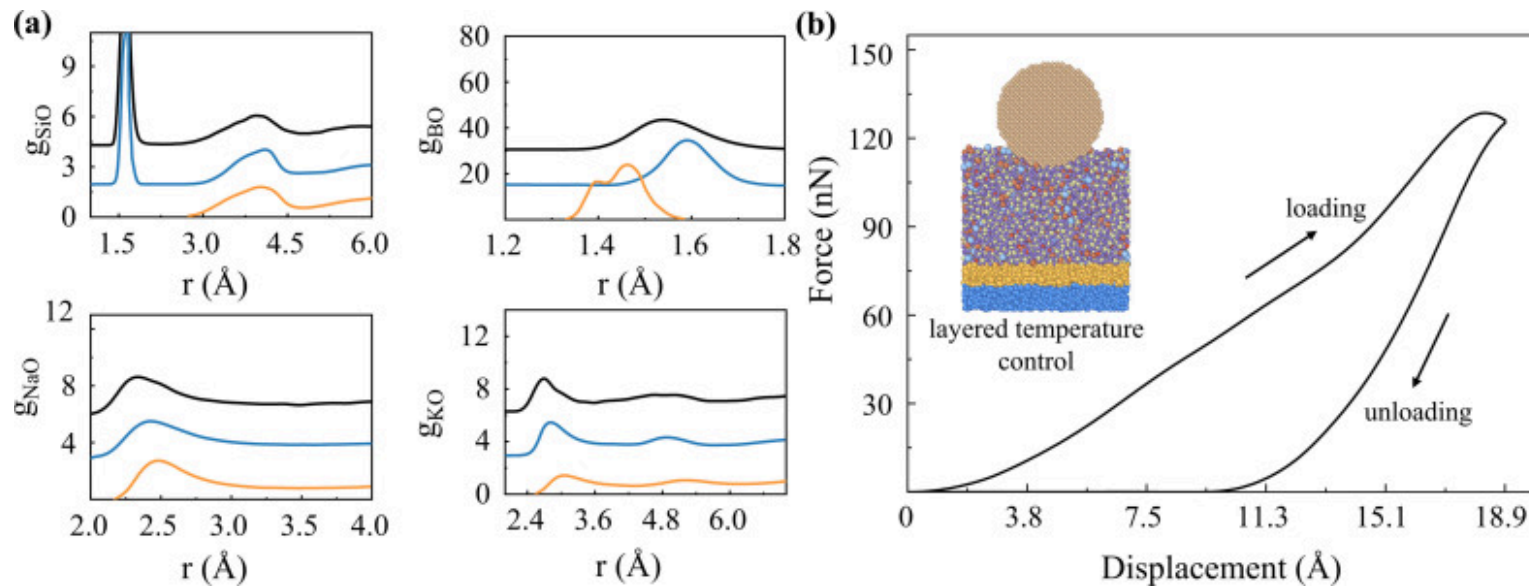
To further evaluate the model's predictive reliability, Fig. 7(c) compares the measured and predicted adhesion forces for the three molds across different molding temperatures, with an overall fitting R^2 of 0.72. The R-squared values for WC, DLC, and AlCrN molds are 0.71, 0.78, and 0.75, respectively, indicating the highest fitting accuracy for DLC and the lowest for WC molds. Fig. 7(d) presents the relative errors of the predicted adhesion forces, showing that the overall average error decreases as the molding temperature increases. The maximum prediction error occurs at 525 °C for DLC mold, possibly due to the small magnitude and limited fluctuation of measured adhesion force. When the temperature rises to 535 °C and beyond, the model overall average prediction error stabilizes around 22.72 %, with a maximum error of 26.06 % at 540 °C. The results show that the adhesion force model developed for WC molds can also reasonably predict adhesion behavior for DLC and AlCrN coatings, confirming its reliability and applicability.

While reducing T_m , F_m , and T^* is necessary to minimize adhesion force as predicted by the model, this requirement is often constrained by permissible production efficiency and machinery precision. However, the formula also suggests solutions to make the molding duration (t) shorter and the strain rates $\dot{\epsilon}$ smaller. In broader sense, Eq. (5) is proven applicable to specific glass-mold pairs, so the adaptability requires more kinds of glass properties.

3.4. Atomistic mechanism of surface adhesion

Understanding how these factors influence adhesion requires examining the atomistic interactions and structural configurations at the mold-glass interface. For the glass structure, Fig. 8(a) shows the Si-O, B-O, Na-O and K-O PDFs with Du's and Wang's potentials [69,95]. The average Si-O distance is around 1.64 Å, which is in good agreement with the experimental values and others [96]. For the B-O PDFs, the peaks are around 1.54 Å, corresponding to the B-O bond length of 3-coordinated B atoms, also close to the experimental 1.47–1.48 Å [97]. For Na-O and K-O PDFs, the simulated result showed minor differences compared to the works by Wang and Du. Therefore, the N-BK7 glass predicts a satisficing short-range structure. Fig. 8(b) shows the force-displacement curve of glass nanoindentation. It can be seen that some elastic deformation of the glass recovers at unloading. The equation proposed by Oliver and Pharr [98] is feasible from the Fig. 8(b) unloading curve. With Poisson's ratio of 0.258,

determined from the test using the IMCE-RFDA device as described before, the calculated elastic modulus of the glass was 69.17 GPa. This value aligns approximately with the experimental result ($E = 82$ GPa), offering a reasonable comparison to the simulation outcomes reported by Wang [69] and Du [95].



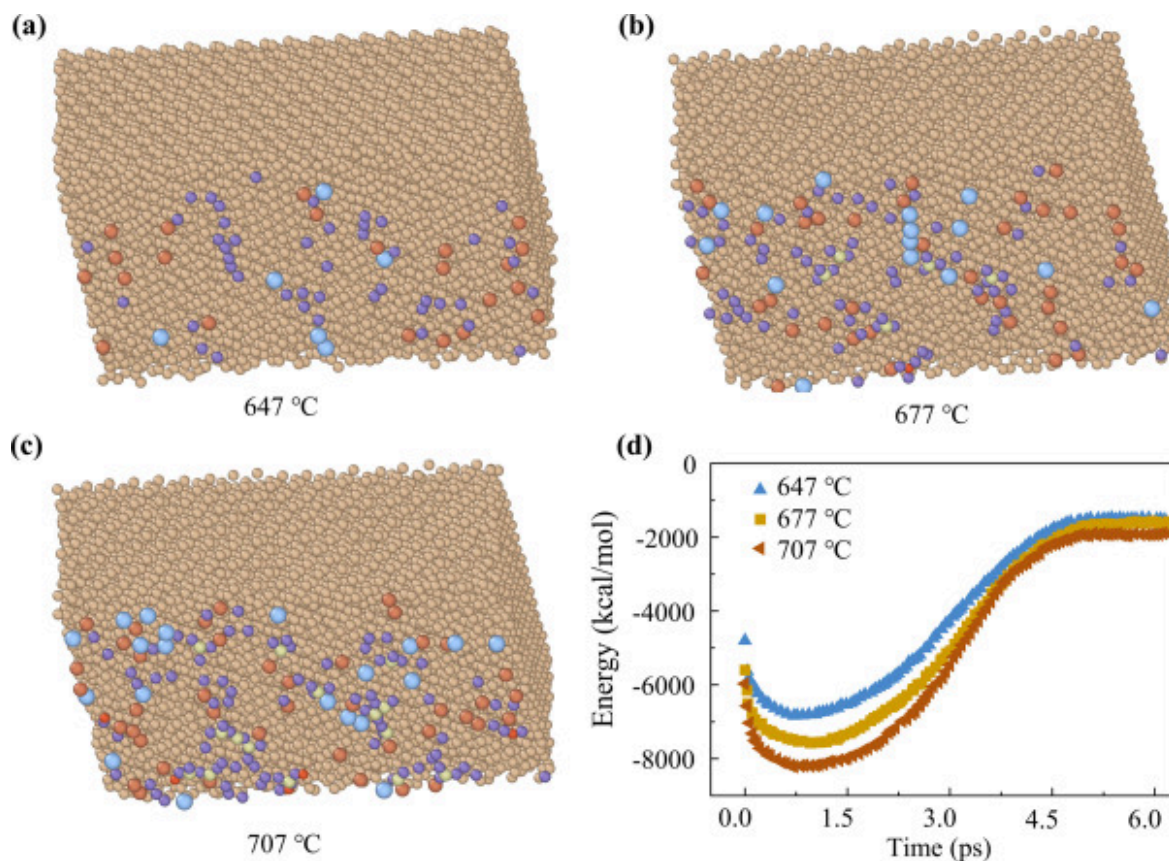
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Fig. 8. Structural validation of simulated N-BK7 glass. (a) Comparison of pair distribution functions (PDFs). (b) Nanoindentation curve during loading and unloading using a diamond indenter.

To investigate the adhesion on the DLC surface after interface separation, it is crucial to examine both the adhered surface and the interaction energy, which varies during the separation process. Fig. 9 illustrates the DLC surface after adhesion at varied molding temperatures of 647, 677, and 707 °C. At all the three temperatures, glass atoms are randomly adhered to the DLC surface without further diffusion into the DLC atoms (Fig. 9(a–c)). The internal breakage in the glass is primarily due to localized atomic separation, where the weakly bonded atoms break their bonds, resulting in cracks at the interface. This adhesion phenomenon, where glass adheres to the mold surface, aligns with the experimental results shown in Fig. 4. The percentage of the adhered glass atoms is 0.5 %, 0.9 %, and 1.2 % for the final upper body consisting of DLC and adhered glass at molding temperatures of 647, 677, and 707 °C, respectively. Correspondingly, the analysis of interface interaction energy (Fig. 9(d)),

indicates that the energy remained negative throughout the separation process, signifying bonding between the glass and mold surfaces. The greater the absolute value of this interaction energy, the stronger the adhesion. At all the three molding temperatures, the interaction energies initially decrease rapidly, reaching a minimum, and then gradually increase to a steady value. The initial rapid decrease in interaction energy (more negative) is likely due to the reduction in atomic distances during pressing, resulting in strong repulsive forces [99]. As demolding begins and atomic distances slightly increase, the interaction energy further decreases within the first 1 ps.



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Fig. 9. Simulated adhesion results at different molding temperatures on DLC mold surfaces. (a-c) Side views of mold surfaces with adhered glass at molding temperatures of 647 °C, 677 °C, and 707 °C, respectively. (d) Interaction energies of the three contact

interfaces during separation.

The minimum interaction energies at molding temperatures of 647, 677, and 707 °C are −6801.75, −7534.52, and −8220.49 kcal/mol, respectively. This indicates that the absolute value of the minimum interaction energy increases with temperature. After 1 ps, the interaction energies gradually increase, eventually converging to higher absolute values at elevated temperatures. This trend aligns with the dimensional analysis and experimental results, suggesting that higher temperatures lead to greater macroscopic adhesion.

4. Conclusions

This study investigated the adhesion mechanisms in precision glass molding by integrating experimental measurements with molecular dynamics simulations. The results revealed that adhesion at the glass-mold interface is significantly influenced by molding parameters, particularly molding temperature, demolding temperature, and molding force, as well as material properties such as glass viscosity and thermodynamic adhesion work. Among these factors, molding temperature was identified as the most critical, as higher temperatures lead to increased adhesion force due to enhanced interface interaction energy and easier glass deformation.

To establish a quantitative relationship between these influencing factors and adhesion force, a dimensionless formulation was developed. This model effectively captured experimental trends and demonstrated satisfactory predictive capabilities across different mold materials. The analysis further revealed that reducing molding temperature and force are most effective ways to mitigate adhesion. Additionally, DLC molds exhibit significantly lower adhesion than WC and AlCrN molds, due to their lower surface energy and reduced interaction with the glass.

By incorporating MD simulations, this study provided atomistic insights into adhesion mechanisms at the interface. The simulations confirmed that adhesion is primarily governed by van der Waals and electrostatic interactions, with higher molding temperatures leading to greater changes in interfacial energy. The simulated separation of the glass-mold interface visualized adhesion, revealing how atomic-scale interactions drive macroscopic adhesion forces observed experimentally. The findings validate the dimensionless formulation while enhancing the understanding of nanoscale adhesion.

Through this research, we hope to provide a theoretical and computational framework for optimizing precision glass molding processes, enabling better control over demolding adhesion and improving mold durability. Future work will extend the

proposed model to a broader range of glass-mold systems and explore novel surface engineering techniques to further enhance the efficiency and reliability of PGM processes.

CRedit authorship contribution statement

Hanhan Zhao: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Software, Visualization. **Liangchi Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization, Project administration. **Asit Kumar Gain:** Writing – review & editing, Investigation. **Zhen Li:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (No. [52293401](#)), the Guangdong Specific Discipline Project (No. 2020ZDZX2006), and the Shenzhen Key Laboratory of Cross Scale Manufacturing Mechanics Project (No. ZDSYS20200810171201007). The research was also supported by the High Level of Special Funds [G030320002](#) and [G03034K003](#) from Southern University of Science and Technology, Shenzhen, China. The authors acknowledge the assistance of SUSTech Core Research Facilities for the SEM observation.

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Data availability

Data will be made available on request.

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